Aurora BP (calibration-based) – Multimodal CNN –BiLSTM-Attention

MODULE 0: CLINICAL PURPOSE AND USE CONTEXT

Model name / version: *Multimodal CNN-BiLSTM-Attention*
Estimation family: [ ] Hemodynamics-derived [**X**] Data-driven BP regression [ ] ABP waveform reconstruction
Calibration paradigm: [ ] Calibration-free [**X**] Calibration-based
Model type: [ ] Machine learning (ML) [**X**] Deep learning (DL) [ ] Hybrid [ ] Other: __________
Input data type: [ ] Handcrafted features [ ] Time series [ ] Images [**X**] Other: *Scalogram (PPG/ECG) and Tabular (demographics)*
Output variable: [**X**] SBP [ ] DBP [ ] MAP... estimated as ΔSBP
Signal modality: [ ] PPG [ ] ECG [ ] BCG [ ] BioZ [ ] IPG [ ] iPPG [**X**] Other: *PPG+ECG+Demographic*
Sensor specification:

- *PPG sensor (MAX30101, 250 S/s),*
- *ECG (ADS1292, ~488–499 S/s),*
- *Tonometric pressure sensor (MS5803-02BA, 200 S/s).*
- *All resampled to 500 S/s common timebase.*

Sensor location:

- *Wrist (tonometric device on radial artery) and proximal forearm (optical/PPG device on anterior arm surface).*
- *ECG via Lead I with dry contacts on wrist + wet electrode on chest.*

Intended use: [] Research use only [ ] Home / ambulatory monitoring [ ] Adjunctive clinical monitoring [**X**] Clinical pilot / prospective evaluation

MODULE 1: DATASET AND COHORT REPORTING

Dataset source: [] Private [**X**] Public: Aurora BP with restricted access (*https://doi.org/10.5281/zenodo.19099166*)
Participants (number of subjects): *463 -483 before data processing*
Recording sessions: *2 in-lab sessions (≥24hrs apart) + ambulatory phase*
Total paired BP samples: *collected over a 24-hour ambulatory protocol*
Population type: [**X**] Healthy [**X**] Hypertensive [ ] ICU [ ] Other: ________
Age (years): *45.6 ± 9.0* ; range *28.0 – 61.0*
Sex: % female *49.5*; % male *50.5*
BMI (kg/m²): NA
Height (cm): *197.8 ± 52.4*
Weight (kg): *67.6 ± 4.0*
Ethnicity / geographic region: *NA*
Medical history / comorbidities reported: [**X**] Yes [ ] No
Medication reported: [**X**] Yes [ ] No;
if yes: class / dose / regimen *(binary yes/no only, no class/dose/regimen detail)*
Known sensitivity: [ ] Motion [ ] Low perfusion [ ] Arrhythmia [ ] Sensor displacement [ ] Skin tone [**X**] Other: *NA*
Skin tone / Fitzpatrick: *Fitzpatrick scale 1–7 collected per participant (no sensitivity analysis reported)* (if PPG-based)
Ambient light conditions: NA (if PPG-based)

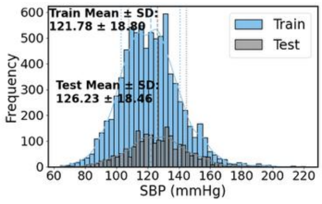
Measurement phases (%):
Ambulatory ___ | In-clinic___ | Dynamic__ | Long-term ___ |Sleep___
BP profiles (%):
Hypotension **0.0** | Normal **36.8** | Elevated **14.8** | Stage 1 **38.4** | Stage 2 **10.0** *(Before processing; after processing: Normal 36.8%, Elevated 14.8%, Stage 1 38.4%, Stage 2 10.0%)*

SBP (mmHg): *128.13 ± 19.74 (before processing); 122.7 ± 18.8 (after processing)*; range: Not reported
DBP (mmHg): *76.06 ± 13.19 (before processing); 74.5 ± 13.0 (after processing)*; range: Not reported
Baseline SBP: *126.12 ± 15.52 (before); 126.1 ± 15.8 (after)*; range: Not reported
Baseline DBP: *77.59 ± 10.01 (before); 77.4 ± 10.2 (after)*; range: Not reported
ΔSBP: *2.07 ± 16.41 (before); 3.4 ± 15.2 (after)* [**X**] Subject-specific [ ] Population-based
ΔDBP: *2.25 ± 11.61 (before); 2.9 ± 11.0 (after)* [**X**] Subject-specific [ ] Population-based

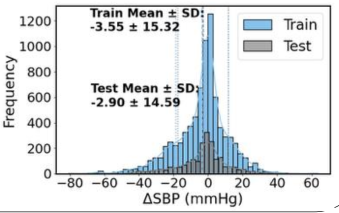
Preprocessing pipeline: *5-sec windowed segmentation; PPG: 4th-order Butterworth bandpass (0.25–10 Hz), baseline correction, beat segmentation; ECG: bior4.4 wavelet denoising; min-max normalization; CWT Morlet scalograms; all signals resampled to 500 S/s. ΔSBP = SBP_baseline − SBP_current*
Signal quality filtering / exclusion criteria: *Optical quality ≥ 0.65; tonometric quality ≥ 0.65; segments with < 5 valid beats discarded; 20 participants excluded during quality check (483 → 463)*
Fiducial points / feature extraction: *Automated deep feature extraction from CWT scalograms (PPG, ECG) via CNN–BiLSTM–Attention; demographics (age, baseline SBP, weight, height, gender) fused via late fusion. No handcrafted features*
Excluded samples after cleaning: *20 participants*
BP distribution before/after cleaning reported: [**X**] Yes [ ] No

BP distribution:

Overall SBP distribution



SBP from calibration



MODULE 2. VALIDATION STRATEGY

Evaluation method: [**X**] Subject-wise hold-out [**X**] Subject-wise CV [ ] LOSO / LOOCV [ ] Record-wise split [ ] Sample-wise split
Data split details: *80:20 subject-level split (370 train, 93 test), stratified to preserve BP category distributions; 5-fold subject-wise cross-validation within training set with no subject overlap between folds*
Unseen-subject test set: [**X**] Yes [ ] No
External validation (independent cohort / site): [**X**] Yes [ ] No (*transfer learning evaluated on SCAI-BP SCI cohort, n=14, via LOSOCV)*
Reference standard: *Cuff-based oscillometric ABP measurement*
Reference device: *Spacelabs Healthcare OnTrak 90227 ABP monitor*

Measurement conditions: [**X**] Resting [**X**] Ambulatory [**X**] Dynamic [ ] Long-term
Body position: *Seated (arm down, arm lap, arm up), supine, standing (arm down, arm up)*
Activity state: *Resting (in-lab postural measurements), ambulatory (free-living 24hr between visits)*
Rest period prior to measurement: *10-min seated rest (oscillometric initial visit); 5-min seated rest (return visit); 5-min rest in-position for each postural change; ≥60 sec between consecutive measurements*
Baseline comparator: *Unimodal PPG-only deep learning model*
Comparator result: *Multimodal model (PPG+ECG+Demographics)*
Minimum acceptable standard met: [**X**] Subject-wise validation

Pathway-specific analyses reported separately in:
[ ] Module 3A (calibration-free)
[**X**] Module 3B (calibration-based)

MODULE 3. PERFORMANCE REPORTING

3B. Calibration-based models

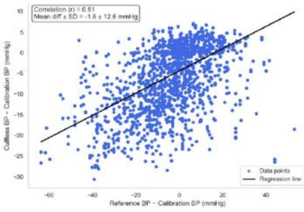
Models evaluated: [**X**] Baseline model [**X**] Proposed / final model
Calibration condition: In-lab baseline sessions
Calibration type: [ ] One-point [**X**] Two-point [ ] Multi-point [ ] Other:_____
Static test: [**X**] Yes [ ] No **Dynamic test:** [**X**] Yes [ ] No
Time since calibration evaluated: [**X**] Yes [ ] No; if yes: **Time interval** (days): *1 day (24 hours)*
Recalibration trigger (if any): *NA*
Performance reporting table: *ΔSBP estimation algorithms (Aurora BP test (hold-out) set, n = 93)*

Condition	ΔSBP±SD (mmHg)	MD±SD (mmHg)	MAD (mmHg)	MAPD (%)	CP5 (%)	CP10 (%)	CP15 (%)	R
Static	-2.85 ± 14.01	-1.15 ± 11.63	8.08 ± 8.43	146.30	50.55	70.38	82.90	0.56
Dynamic	-2.43 ± 14.15	-1.84 ±11.80	8.61 ± 8.28	154.64	44.92	67.63	81.95	0.55
Time after calibration	0.09 ± 12.39	-2.32 ± 10.31	7.77 ± 7.15	160.12	43.92	72.95	72.95	0.55

Baseline evaluation table:

Condition	ΔSBP±SD (mmHg)	MD±SD (mmHg)	MAD (mmHg)	MAPD (%)	CP5 (%)	CP10 (%)	CP15 (%)	R
Static	-	-	10.90 ± 13.25	-	-	-	-	-
Dynamic	-	-	14.48 ± 17.66	-	-	-	-	-
Time after calibration	-	-	-	-	-	-	-	-

Scatter plot ΔSBP



Bland -Altman Plot ΔSBP